

Exp-4. Working with Hive on the following

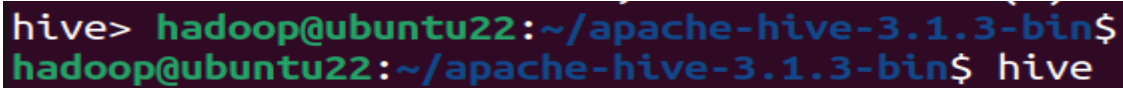
i) HiveQL

Date: 15-02-2025

hadoop@ubuntu22:~\$ cd \$HIVE_HOME

hadoop@ubuntu22:~/apache-hive-3.1.3-bin\$ hive

hive>



```
hive> hadoop@ubuntu22:~/apache-hive-3.1.3-bin$  
hadoop@ubuntu22:~/apache-hive-3.1.3-bin$ hive
```

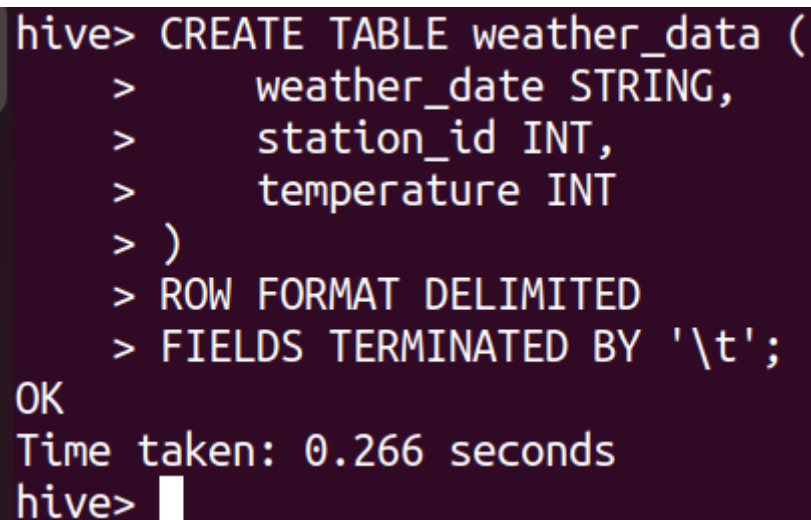
1. Creating a Hive Table

```
hive> CREATE TABLE weather_data (  
  >   weather_date STRING,  
  >   station_id INT,  
  >   temperature INT  
  > )  
  > ROW FORMAT DELIMITED  
  > FIELDS TERMINATED BY '\t';
```

OK

Time taken: 0.266 seconds

hive>



```
hive> CREATE TABLE weather_data (  
  >   weather_date STRING,  
  >   station_id INT,  
  >   temperature INT  
  > )  
  > ROW FORMAT DELIMITED  
  > FIELDS TERMINATED BY '\t';  
OK  
Time taken: 0.266 seconds  
hive>
```

```
hive> show tables;
OK
temperature
temperature_data
weather_data
Time taken: 0.183 seconds, Fetched: 3 row(s)
```

```
hive>
```

```
hive> describe weather_data;
OK
weather_date          string
station_id            int
temperature            int
Time taken: 0.203 seconds, Fetched: 3 row(s)
```

```
hive>
```

```
hive> show tables;
OK
temperature
temperature_data
weather_data
Time taken: 0.183 seconds, Fetched: 3 row(s)
hive> describe weather_data;
OK
weather_date          string
station_id            int
temperature            int
Time taken: 0.203 seconds, Fetched: 3 row(s)
hive>
```

2. Loading Data into the Table

Input File: weather_sample.txt

10-01-1990	123112	10
14-02-1991	283901	11
10-03-1990	381920	15
10-01-1991	302918	22
12-02-1990	384902	9
10-01-1991	123112	11
14-02-1990	283901	12
10-03-1991	381920	16
10-01-1990	302918	23
12-02-1991	384902	10
10-01-1993	123112	11
14-02-1994	283901	12
10-03-1993	381920	16
10-01-1994	302918	23
12-02-1991	384902	10
10-01-1991	123112	11
14-02-1990	283901	12
10-03-1991	381920	16
10-01-1990	302918	23
12-02-1991	384902	10

```
hive> LOAD DATA LOCAL INPATH  
'/home/hadoop/Desktop/RMMBDALAB/Exp-4.Hive/weather_sample.txt'  
INTO TABLE weather_data;
```

Loading data to table default.weather_data
OK
Time taken: 1.061 seconds

```
hive> LOAD DATA LOCAL INPATH '/home/hadoop/Desktop/RMMBDALAB/Exp-4.Hive/weather_sample.txt' INTO TA  
BLE weather_data;  
Loading data to table default.weather_data  
OK  
Time taken: 1.061 seconds
```

3. Selecting Data from Table

```
hive> SELECT * FROM weather_data;
```

OK

10-01-1990	123112	10
14-02-1991	283901	11
10-03-1990	381920	15
10-01-1991	302918	22
12-02-1990	384902	9
10-01-1991	123112	11
14-02-1990	283901	12
10-03-1991	381920	16
10-01-1990	302918	23
12-02-1991	384902	10
10-01-1993	123112	11
14-02-1994	283901	12
10-03-1993	381920	16
10-01-1994	302918	23
12-02-1991	384902	10
10-01-1991	123112	11
14-02-1990	283901	12
10-03-1991	381920	16
10-01-1990	302918	23
12-02-1991	384902	10

Time taken: 1.152 seconds, Fetched: 20 row(s)

```
hive> SELECT * FROM weather_data;
```

OK

10-01-1990	123112	10
14-02-1991	283901	11
10-03-1990	381920	15
10-01-1991	302918	22
12-02-1990	384902	9
10-01-1991	123112	11
14-02-1990	283901	12
10-03-1991	381920	16
10-01-1990	302918	23
12-02-1991	384902	10
10-01-1993	123112	11
14-02-1994	283901	12
10-03-1993	381920	16
10-01-1994	302918	23
12-02-1991	384902	10
10-01-1991	123112	11
14-02-1990	283901	12
10-03-1991	381920	16
10-01-1990	302918	23
12-02-1991	384902	10

Time taken: 1.152 seconds, Fetched: 20 row(s)

```
hive> █
```

```
hive> SELECT * FROM weather_data LIMIT 5;
```

OK

10-01-1990	123112	10
14-02-1991	283901	11
10-03-1990	381920	15
10-01-1991	302918	22
12-02-1990	384902	9

Time taken: 0.299 seconds, Fetched: 5 row(s)

```
hive>
```

```
hive> SELECT * FROM weather_data LIMIT 5;
```

OK

10-01-1990	123112	10
14-02-1991	283901	11
10-03-1990	381920	15
10-01-1991	302918	22
12-02-1990	384902	9

Time taken: 0.299 seconds, Fetched: 5 row(s)

```
hive> █
```

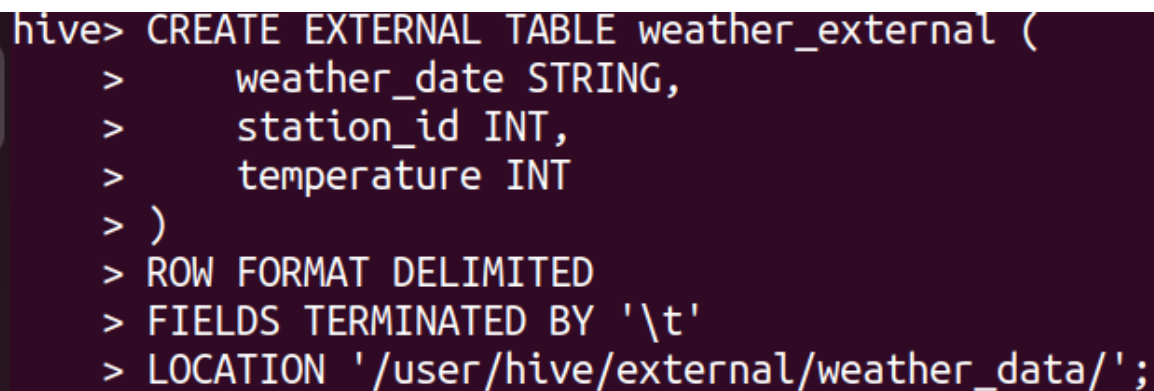
4. Creating an External Table

```
hive> CREATE EXTERNAL TABLE weather_external (  
  >     weather_date STRING,  
  >     station_id INT,  
  >     temperature INT  
  > )  
  > ROW FORMAT DELIMITED  
  > FIELDS TERMINATED BY '\t'  
  > LOCATION '/user/hive/external/weather_data/';
```

OK

Time taken: 0.181 seconds

hive>

A screenshot of a terminal window with a dark purple background. It shows the same Hive command as above: 'hive> CREATE EXTERNAL TABLE weather_external (weather_date STRING, station_id INT, temperature INT) ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t' LOCATION '/user/hive/external/weather_data/';'. The text is in a light-colored monospace font.

```
hive> CREATE EXTERNAL TABLE weather_external (  
  >     weather_date STRING,  
  >     station_id INT,  
  >     temperature INT  
  > )  
  > ROW FORMAT DELIMITED  
  > FIELDS TERMINATED BY '\t'  
  > LOCATION '/user/hive/external/weather_data/';
```

OK

Time taken: 0.181 seconds

hive>

```
hive> show tables;
```

OK

temperature

temperature_data

weather_data

weather_external

Time taken: 0.07 seconds, Fetched: 4 row(s)

hive>

```
hive> describe weather_external;
```

OK

weather_date	string
--------------	--------

station_id	int
------------	-----

temperature	int
-------------	-----

Time taken: 0.133 seconds, Fetched: 3 row(s)

hive>

```
hive> show tables;
OK
temperature
temperature_data
weather_data
weather_external
Time taken: 0.07 seconds, Fetched: 4 row(s)
hive> describe weather_external;
OK
weather_date          string
station_id            int
temperature            int
Time taken: 0.133 seconds, Fetched: 3 row(s)
hive> █
```

5. Partitioning and Bucketing

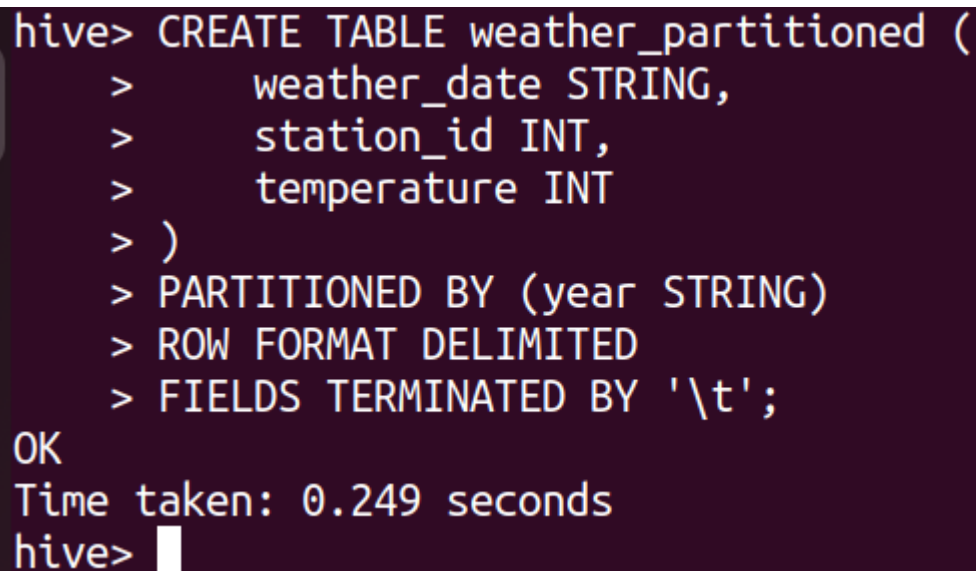
Partitioning by Year:

```
hive> CREATE TABLE weather_partitioned (  
  >   weather_date STRING,  
  >   station_id INT,  
  >   temperature INT  
  > )  
  > PARTITIONED BY (year STRING)  
  > ROW FORMAT DELIMITED  
  > FIELDS TERMINATED BY '\t';
```

OK

Time taken: 0.249 seconds

hive>

A screenshot of a terminal window with a dark purple background. It shows the same Hive SQL command as above, followed by the response 'OK', 'Time taken: 0.249 seconds', and a new prompt 'hive>' with a white cursor.

```
hive> CREATE TABLE weather_partitioned (  
  >   weather_date STRING,  
  >   station_id INT,  
  >   temperature INT  
  > )  
  > PARTITIONED BY (year STRING)  
  > ROW FORMAT DELIMITED  
  > FIELDS TERMINATED BY '\t';  
OK  
Time taken: 0.249 seconds  
hive> █
```

```
hive> ALTER TABLE weather_partitioned ADD PARTITION  
(year='1990');
```

OK

Time taken: 0.221 seconds

hive>

Output:

Partitioned table created and partition added.

```
hive> show tables;
OK
temperature
temperature_data
weather_data
weather_external
weather_partitioned
Time taken: 0.104 seconds, Fetched: 5 row(s)
hive>
```

```
hive> describe weather_partitioned;
OK
weather_date          string
station_id            int
temperature            int
year                  string
# Partition Information
# col_name            data_type            comment
year                  string
```

Time taken: 0.349 seconds, Fetched: 8 row(s)

```
hive>
```

```
hive> show tables;
OK
temperature
temperature_data
weather_data
weather_external
weather_partitioned
Time taken: 0.104 seconds, Fetched: 5 row(s)
hive> describe weather_partitioned;
OK
weather_date          string
station_id            int
temperature            int
year                  string

# Partition Information
# col_name            data_type            c
year                  string
Time taken: 0.349 seconds, Fetched: 8 row(s)
hive>
```

Bucketing by Station ID:

```
hive> CREATE TABLE weather_bucketed (  
  >     weather_date STRING,  
  >     station_id INT,  
  >     temperature INT  
  > )  
  > CLUSTERED BY (station_id) INTO 4 BUCKETS  
  > ROW FORMAT DELIMITED  
  > FIELDS TERMINATED BY '\t';
```

OK

Time taken: 0.292 seconds

hive>

```
hive> show tables;
```

OK

```
temperature  
temperature_data  
weather_bucketed  
weather_data  
weather_external  
weather_partitioned
```

Time taken: 0.087 seconds, Fetched: 6 row(s)

```
hive> CREATE TABLE weather_bucketed (  
  >     weather_date STRING,  
  >     station_id INT,  
  >     temperature INT  
  > )  
  > CLUSTERED BY (station_id) INTO 4 BUCKETS  
  > ROW FORMAT DELIMITED  
  > FIELDS TERMINATED BY '\t';  
OK  
Time taken: 0.292 seconds  
hive> show tables;  
OK  
temperature  
temperature_data  
weather_bucketed  
weather_data  
weather_external  
weather_partitioned  
Time taken: 0.087 seconds, Fetched: 6 row(s)  
hive>
```

```
hive> describe weather_bucketed;  
OK  
weather_date          string  
station_id            int  
temperature            int  
Time taken: 0.255 seconds, Fetched: 3 row(s)  
hive>
```

```
hive> describe weather_bucketed;  
OK  
weather_date          string  
station_id            int  
temperature            int  
Time taken: 0.255 seconds, Fetched: 3 row(s)  
hive>
```

Output:

Bucketed table created.

6. Altering a Table

```
hive> ALTER TABLE weather_data ADD COLUMNS (humidity INT);
```

OK

Time taken: 0.208 seconds

```
hive> describe weather_data;
```

OK

weather_date	string
station_id	int
temperature	int
humidity	int

Time taken: 0.111 seconds, Fetched: 4 row(s)

hive>

Output:

Column humidity added.

```
hive> ALTER TABLE weather_data ADD COLUMNS (humidity INT);
```

OK

Time taken: 0.208 seconds

```
hive> describe weather_data;
```

OK

weather_date	string
station_id	int
temperature	int
humidity	int

Time taken: 0.111 seconds, Fetched: 4 row(s)

```
hive> █
```

7. Dropping a Table

```
DROP TABLE weather_data;
```

Output:

Table weather_data dropped.

```
hive> DROP TABLE weather_data;
```

OK

Time taken: 0.229 seconds

```
hive> show tables;
```

OK

temperature

temperature_data

weather_bucketed

weather_external

weather_partitioned

Time taken: 0.026 seconds, Fetched: 5 row(s)

```
hive>
```

```
hive> describe weather_data;
```

FAILED: SemanticException [Error 10001]: Table not found weather_data

```
hive>
```

```
hive> DROP TABLE weather_data;
OK
Time taken: 0.229 seconds
hive> show tables;
OK
temperature
temperature_data
weather_bucketed
weather_external
weather_partitioned
Time taken: 0.026 seconds, Fetched: 5 row(s)
hive> describe weather_data;
FAILED: SemanticException [Error 10001]: Table not found weather_data
hive>
```

8. Querying Data

Finding Maximum Temperature

```
SELECT MAX(temperature) FROM weather_data;
```

Output:

```
23
```

Finding Minimum Temperature

```
SELECT MIN(temperature) FROM weather_data;
```

Output:

```
9
```

9. Sorting and Aggregation

Sorting by Temperature (Descending)

```
SELECT * FROM weather_data ORDER BY temperature DESC;
```

Output:

Records sorted in descending order.

Aggregating Average Temperature by Year

```
SELECT substr(date, 7, 4) AS year, AVG(temperature) FROM weather_data GROUP BY year;
```

Output:

1990	14.8
1991	13.5
1993	13.5
1994	17.5

10. Joins

Creating another Table

```
hive> CREATE TABLE station_info (station_id INT, location STRING) ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t';
```

OK

Time taken: 0.27 seconds

```
hive> LOAD DATA LOCAL INPATH '/home/hadoop/Desktop/rmm/stationdata.txt' INTO TABLE station_info;
```

Loading data to table default.station_info

OK

Time taken: 0.749 seconds

```
hive> select * from station_info;
```

OK

123112	NYC
283901	LA
381920	Chicago
302918	Houston
384902	Dallas

Time taken: 0.293 seconds, Fetched: 10 row(s)

```
hive> select * from weather_data;
```

OK

10-01-1990	123112	10
14-02-1991	283901	11
10-03-1990	381920	15
10-01-1991	302918	22
12-02-1990	384902	9

10-01-1991	123112	11
14-02-1990	283901	12
10-03-1991	381920	16
10-01-1990	302918	23
12-02-1991	384902	10
10-01-1993	123112	11
14-02-1994	283901	12
10-03-1993	381920	16
10-01-1994	302918	23
12-02-1991	384902	10
10-01-1991	123112	11
14-02-1990	283901	12
10-03-1991	381920	16
10-01-1990	302918	23
12-02-1991	384902	10

Time taken: 0.152 seconds, Fetched: 20 row(s)

hive> select * from station_info;

OK

123112	NYC
283901	LA
381920	Chicago
302918	Houston
384902	Dallas

Time taken: 0.196 seconds, Fetched: 10 row(s)

Joining weather_data and station_info

```
hive> SELECT w.weather_date, w.temperature, s.location FROM weather_data w  
JOIN station_info s ON w.station_id = s.station_id;
```

```
hive> SELECT w.weather_date, w.temperature, s.location FROM weather_data w JOIN station_info s ON w  
.station_id = s.station_id;
```

Output:

OK

10-01-1990	10	NYC
14-02-1991	11	LA
10-03-1990	15	Chicago
10-01-1991	22	Houston
12-02-1990	9	Dallas
10-01-1991	11	NYC
14-02-1990	12	LA
10-03-1991	16	Chicago
10-01-1990	23	Houston
12-02-1991	10	Dallas
10-01-1993	11	NYC
14-02-1994	12	LA
10-03-1993	16	Chicago
10-01-1994	23	Houston
12-02-1991	10	Dallas
10-01-1991	11	NYC
14-02-1990	12	LA
10-03-1991	16	Chicago
10-01-1990	23	Houston
12-02-1991	10	Dallas

Time taken: 15.671 seconds, Fetched: 20 row(s)

```
OK
10-01-1990      10      NYC
14-02-1991      11      LA
10-03-1990      15      Chicago
10-01-1991      22      Houston
12-02-1990       9      Dallas
10-01-1991      11      NYC
14-02-1990      12      LA
10-03-1991      16      Chicago
10-01-1990      23      Houston
12-02-1991      10      Dallas
10-01-1993      11      NYC
14-02-1994      12      LA
10-03-1993      16      Chicago
10-01-1994      23      Houston
12-02-1991      10      Dallas
10-01-1991      11      NYC
14-02-1990      12      LA
10-03-1991      16      Chicago
10-01-1990      23      Houston
12-02-1991      10      Dallas
Time taken: 15.671 seconds, Fetched: 20 row(s)
hive> █
```

11. Subqueries

```
SELECT * FROM weather_data WHERE temperature = (SELECT MAX(temperature) FROM weather_data);
```

Output:

Records with the highest temperature.

```
hive> SELECT * FROM weather_data WHERE temperature = 12;
```

OK

```
14-02-1990    283901 12
```

```
14-02-1994    283901 12
```

```
14-02-1990    283901 12
```

Time taken: 0.222 seconds, Fetched: 3 row(s)

```
hive>
```

```
hive> SELECT * FROM weather_data WHERE temperature = 12;
```

OK

```
14-02-1990    283901 12
```

```
14-02-1994    283901 12
```

```
14-02-1990    283901 12
```

Time taken: 0.222 seconds, Fetched: 3 row(s)

```
hive> █
```

12. Views

```
CREATE VIEW temp_summary AS  
SELECT station_id, AVG(temperature) AS avg_temp FROM weather_data GROUP BY  
station_id;
```

```
SELECT * FROM temp_summary;
```

Output:

```
123112  10.5  
283901  11.6  
381920  15.5  
...
```